



Problems (Differential Equations)

1. Find the solution $y(x)$ to the following initial value problem: $\frac{dy}{dx} = x^2 + e^{3x} + 1$, $y(0)=0$
2. Find the solution $y(x)$ to the following initial value problem: $\frac{dy}{dx} = \sqrt{1 - y^2}$, $y(0)=1$
3. An amount of $A(0)$ USD is invested against yearly interest of $r\%$. Give the expression for $A(t)$, the value of the investment in USD after t years if the interest is compounded continuously by writing down the differential equation that A satisfies and solving it.
4. In a certain culture of bacteria, the number of bacteria increased tenfold in 10 hours. Assuming natural growth, how long did it take for their number to double?
5. The rate at which a student learns new material is proportional to the difference between a maximum, M , and the amount she already knows at time t , $A(t)$. This is called a learning curve.
 - a. Write a differential equation to model the learning curve described.
 - b. Solve the differential equation you created in part (a).
 - c. If took a student 100 hours to learn 50% of the material in Quant Finance and she would like to know 75% to get a grade B, how much longer she should study? You may assume that the student began knowing none of the material and that the maximum she might achieve is 100%.
6. Solve the following differential equation: $x \frac{dy}{dx} + 3y = 8x^5$
7. Show that the solution to the equation $x^2 y'' - 2xy' + 2y = 4x^3$ is given by $y = Ax + Bx^2 + 2x^3$